1 Physics-aware encoding operator and its normal operator

We reconstruct complex images x on a Cartesian grid $\Omega \subset \mathbb{R}^3$ from multi-coil spiral data acquired without density compensation. The data model must include everything the sequence *determines*: off-resonance, the concomitant (Maxwell) fields of the *whole* gradient history, gradient nonlinearity (GNL), and cardiac motion between the readouts of one frame. This section derives the operator A that does so at the cost of $L \approx 10$ FFT pairs per coil (Section 1.3), and the algebra of its normal operator $\mathsf{N} = A^{\mathsf{H}}A$ (Section 1.4), which explains why the correction terms are small, exact, and cheap, and which fixes how the operator enters the preconditioned plug-and-play iteration (Section 1.5). The operators are model-exact up to the controlled truncations stated below (time-segmentation [3, 4], a Taylor series in the small per-readout remainder); none of them uses density compensation, which is replaced by preconditioning and coil-noise whitening.

1.1 Signal model and the phase budget

Let $\boldsymbol{r}$ be the *true scanner position* of a voxel (after the GNL warp, including the FOV offset from isocentre), S_c the coil sensitivities, and $j = 1, \dots, M$ the samples of one set (one frame $\times$ one encode). With $\boldsymbol{k}_j$ the nominal trajectory (rad/m) and τ_j the time since excitation,

$$y_{cj} \;=\; \sum_{\boldsymbol{r}\in\Omega} S_c(\boldsymbol{r})\, x(\boldsymbol{r})\, \mathrm{e}^{-\mathrm{i}\Theta_j(\boldsymbol{r})} + n_{cj}, \qquad \Theta_j(\boldsymbol{r}) \;=\; \boldsymbol{k}_j\cdot\boldsymbol{r} \;+\; \omega(\boldsymbol{r})\,\tau_j \;+\; \sum_{q=1}^{4} w_{jq}\, m_q(\boldsymbol{r}), \tag{1}$$

where ω is the off-resonance map (rad/s), $m = (z^2,\ x^2{+}y^2,\ xz,\ yz)$ are the concomitant monomials in *physical* coordinates, and w_{jq} are sequence-determined coefficients (below). Noise is complex Gaussian with coil covariance Ψ; we whiten data and maps with $\Psi^{-1/2}$ and drop the tilde. T_2^* decay is carried by x (and by ω if one admits $\operatorname{Im}\omega \neq 0$); it is a smooth function of τ and costs no extra segments.

Concomitant coefficients. A linear gradient $\boldsymbol{G} = (G_x, G_y, G_z)$ that satisfies Maxwell's equations has transverse components [2, 1]. The field with $\nabla\cdot\boldsymbol{B} = 0$, $\nabla\times\boldsymbol{B} = 0$ is

$$\boldsymbol{B} \;=\; (B_0 + \boldsymbol{G}\cdot\boldsymbol{r})\,\hat{z} \;+\; (G_x z - \tfrac{1}{2}G_z x)\,\hat{x} \;+\; (G_y z - \tfrac{1}{2}G_z y)\,\hat{y}, \tag{2}$$

(divergence: $G_z - \frac{1}{2}G_z - \frac{1}{2}G_z = 0$; curl: all components vanish). Its magnitude is, to leading order in G/B_0, $|\boldsymbol{B}| \approx B_z + B_\perp^2/(2B_0)$, so the concomitant field is

$$B_c \;=\; \frac{(G_x z - \frac{1}{2}G_z x)^2 + (G_y z - \frac{1}{2}G_z y)^2}{2B_0} \;=\; \frac{G_x^2 + G_y^2}{2B_0}\, z^2 + \frac{G_z^2}{8B_0}(x^2{+}y^2) - \frac{G_x G_z}{2B_0}\, xz - \frac{G_y G_z}{2B_0}\, yz. \tag{3}$$

The phase accumulated by a spin at $\boldsymbol{r}$ from excitation to sample j is therefore $\sum_q w_{jq} m_q(\boldsymbol{r})$ with

$$\begin{aligned} w_{j1} &= \tfrac{\gamma}{2B_0}\textstyle\int_0^{t_j} (G_x^2 + G_y^2)\,\mathrm{d}t, \quad & w_{j2} &= \tfrac{\gamma}{8B_0}\textstyle\int_0^{t_j} G_z^2\,\mathrm{d}t, \\ w_{j3} &= -\tfrac{\gamma}{2B_0}\textstyle\int_0^{t_j} G_x G_z\,\mathrm{d}t, & w_{j4} &= -\tfrac{\gamma}{2B_0}\textstyle\int_0^{t_j} G_y G_z\,\mathrm{d}t \end{aligned} \tag{4}$$

(overall sign absorbed to match Equation (1)). Equivalently $\sum_q w_{jq} m_q = \frac{\gamma}{2B_0}\langle W_j, S(\boldsymbol{r})\rangle$ with the second-moment matrix $W_j = \int_0^{t_j} \boldsymbol{G}\boldsymbol{G}^\top \mathrm{d}t$ and $S = M_{\boldsymbol{r}}^\top M_{\boldsymbol{r}}$, $M_{\boldsymbol{r}} = \left(\begin{smallmatrix} z & 0 & -x/2 \\ 0 & z & -y/2 \end{smallmatrix}\right)$. Three properties are used below.

1. *History dependence.* W_j integrates from excitation, so the velocity-encoding bipolars, the k_z blip and the readout all contribute. The part of the bipolars that is constant over the readout is the per-encode Maxwell velocity offset; it is a time-independent phase $\varphi_e(\boldsymbol{r}) = \sum_q w^{\mathrm{enc}}_{e,q} m_q(\boldsymbol{r})$ and enters the model with no special treatment (it is inside the segment diagonals).

2. *Frame.* $\boldsymbol{G}$ is the *physical* gradient: the logical→physical rotation and the FOV offset are applied before Equation (4). In logical voxel coordinates $\boldsymbol{\rho}$ each m_q is a general quadratic polynomial (with linear and constant parts from the offset).

3. *Accuracy.* Equation (3) neglects $O(B_\perp^4/B_0^3)$; the relative error is $\sim 2{\cdot}10^{-4}$ at 3 T, 20 mT/m, 10 cm (checked against a direct evaluation of $|\boldsymbol{B}|$). The magnitude of the coefficients matters: all rank, order and accuracy statements below scale with $\Phi_{\max}$, which is set by the z^2 coefficient $(G_x^2{+}G_y^2)/(2B_0)$ and the cross-term signs.

Size of the effect. For the simulated 3 T spiral 4D-flow sequence (80 mT/m, 200 T/m/s, 36 cm in-plane FOV, 56 mm slab, 4.4 ms readout, FOV centre 3 cm off isocentre) the concomitant phase over all samples and the FOV, expressed in the FOV-centred logical frame $\boldsymbol{\rho}$ as $\phi_0 + \boldsymbol{b}{\cdot}\boldsymbol{\rho} + \boldsymbol{\rho}^\top C\boldsymbol{\rho}$, is [measured]:

part	max over samples and FOV
constant ϕ_0	0.08 rad
linear $\boldsymbol{b}{\cdot}\boldsymbol{\rho}$ (= k-space shift of ≤ 0.29 cells)	0.96 rad
quadratic $\boldsymbol{\rho}^\top C\boldsymbol{\rho}$ (FOV corners)	3.1 rad

The constant is a sample weight, the linear part is an *exact* shift of the sample location ($\boldsymbol{k}'_j = \boldsymbol{k}_j + \boldsymbol{b}_j$), and only the quadratic part needs operator work. At 3 rad neither a global first-order nor a global second-order expansion of $\mathrm{e}^{-\mathrm{i}\Theta}$ is accurate, which motivates splitting the phase into a part carried exactly by segments and a small per-sample remainder.

1.2 Splitting the phase: time segments plus a per-sample deviation

Group the readouts of a set by ADC index n (same τ_n in every readout) and let $\bar{w}_q(\tau_n)$ be the mean of w_{jq} over the readouts of the set (or, better, over all frames of the encode) at that ADC index. Write

$$\Theta_j(\boldsymbol{r}) = \boldsymbol{k}_j{\cdot}\boldsymbol{r} + \underbrace{\omega(\boldsymbol{r})\tau_j + \sum_q \bar{w}_q(\tau_j) m_q(\boldsymbol{r})}_{\psi(\boldsymbol{r},\tau_j)} + \underbrace{\sum_q \Delta w_{jq}\, m_q(\boldsymbol{r})}_{\Delta_j(\boldsymbol{r})}, \qquad \Delta w_{jq} = w_{jq} - \bar{w}_q(\tau_j). \tag{5}$$

- ψ is a function of $\boldsymbol{r}$ and the single scalar τ. It holds the B0 map, the readout-time-dependent concomitant phase and the bipolar offset. It may be *any* spatial function (post-shim field map, GNL-warped positions, FOV offset): no polynomial assumption.

- Δ_j is the part that depends on the individual readout (rotation of the spiral about k_z or in 3D, blip height). It is small and *quadratic*: least-squares fit each m_q over the object support to $\theta_q \cdot [1, \boldsymbol{r}, r_p r_q]$ and set

$$\Delta_j(\boldsymbol{r}) \approx \phi_{0j} + \boldsymbol{b}_j{\cdot}\boldsymbol{r} + \boldsymbol{r}^\top \Delta C_j\, \boldsymbol{r}. \tag{6}$$

  The fit residual $\sum_q \max|\Delta w_{jq}| \max|m_q - \mathrm{fit}|$ is recorded as the phase the model leaves out.

Time segments. Choose segment times $\tau_1 < \cdots < \tau_L$ (uniform over the readout) and diagonals $D_\ell(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau_\ell)}$. For each sample time τ the interpolation coefficients are the least-squares solution over the object support Ω_s,

$$\begin{aligned} \boldsymbol{a}(\tau) = \arg\min_{\boldsymbol{a}\in\mathbb{C}^L} \sum_{\boldsymbol{r}\in\Omega_s} \left|\mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau)} - \sum_\ell a_\ell D_\ell(\boldsymbol{r})\right|^2 \iff G\,\boldsymbol{a}(\tau) = \boldsymbol{\beta}(\tau), \\ G_{\ell\ell'} = \sum_{\boldsymbol{r}} \overline{D_\ell} D_{\ell'}, \qquad \beta_\ell(\tau) = \sum_{\boldsymbol{r}} \overline{D_\ell(\boldsymbol{r})}\mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau)}. \end{aligned} \tag{7}$$

This is the least-squares variant of the time-segmentation idea of Sutton–Noll–Fessler [3], with the support-weighted ℓ_2 criterion; the $a_\ell(\tau)$ are computed once per scan (a small $L \times L$ solve per ADC index).

How many segments? The rank of $\mathrm{e}^{-\mathrm{i}\psi}$ is set by the *range* of ψ, not by resolution [measured, CUR rank on body voxels]:

phase in ψ	L at 10^{-2}	L at $2{\cdot}10^{-3}$
B0 only	8	9
quadratic concomitant only	4	6
B0 + quadratic concomitant, one global decomposition	15	22

The joint rank is roughly the *sum* when one global decomposition has to represent both, but the segment count with the split Equation (5) is set by B0 alone: the τ-dependent concomitant part rides in the same diagonals, and the readout-dependent remainder is handled per sample (Section 1.3). With segments carrying both and the first-order ΔC correction, $L = 8, 10, 12$ reach $3.2{\cdot}10^{-3}$, $8.9{\cdot}10^{-4}$, $4.3{\cdot}10^{-4}$ (versus a plateau at $1.3{\cdot}10^{-2}$ without the correction) [measured]. The B0 rank grows linearly with readout length, $L \approx 2\,\Delta f\, T_{\mathrm{ro}} + 4$ with Δf the 1–99 % off-resonance span (here 0.7 kHz; 4.4 ms $\Rightarrow L \approx 10$). Smoothing B0 into the trajectory (linear part $\to$ time-dependent k-shift, quadratic part $\to C$) does *not* reduce L: post-shim B0 is local susceptibility structure, not a smooth field.

1.3 The segmented physics NUFFT

Forward operator. Insert Equation (5)–Equation (6) and the segment expansion into Equation (1). With $\boldsymbol{k}'_j = \boldsymbol{k}_j + \boldsymbol{b}_j$ and $v_{c\ell}(\boldsymbol{r}) = S_c(\boldsymbol{r})x(\boldsymbol{r})D_\ell(\boldsymbol{r})$,

$$(Ax)_{cj} \;\approx\; \mathrm{e}^{-\mathrm{i}\phi_{0j}} \sum_{\ell=1}^{L} a_\ell(\tau_j) \sum_{\boldsymbol{r}} v_{c\ell}(\boldsymbol{r})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{k}'_j\cdot\boldsymbol{r}}\, \mathrm{e}^{-\mathrm{i}\,\boldsymbol{r}^\top \Delta C_j \boldsymbol{r}}. \tag{8}$$

Let $V_{c\ell}(\boldsymbol{k}) = \sum_{\boldsymbol{r}} v_{c\ell}(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$. Because $\boldsymbol{r}^\top\Delta C\boldsymbol{r}$ is quadratic, the chirp is a polynomial *differential operator in* $\boldsymbol{k}$: from $\partial_{k_p} V = \sum(-\mathrm{i}r_p)v\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$,

$$\mathcal{F}\{r_p r_q\, v\} = -\partial_p\partial_q V, \qquad \mathcal{F}\{r_p r_q r_r r_s\, v\} = \partial_p\partial_q\partial_r\partial_s V, \tag{9}$$

and expanding $\mathrm{e}^{-\mathrm{i}u} = 1 - \mathrm{i}u - \frac{1}{2}u^2 + \dots$ with $u = \boldsymbol{r}^\top\Delta C\boldsymbol{r}$,

$$\sum_{\boldsymbol{r}} v(\boldsymbol{r})\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\mathrm{e}^{-\mathrm{i}\boldsymbol{r}^\top\Delta C\boldsymbol{r}} \;=\; V(\boldsymbol{k}) + \mathrm{i}\sum_{pq}\Delta C_{pq}\,\partial_p\partial_q V(\boldsymbol{k}) - \tfrac{1}{2}\sum_{pqrs}\Delta C_{pq}\Delta C_{rs}\,\partial_p\partial_q\partial_r\partial_s V(\boldsymbol{k}) + \dots \tag{10}$$

(checked numerically against the exact chirped sum). No extra FFT is needed: the NUFFT evaluates V at $\boldsymbol{k}'_j$ by interpolating the oversampled FFT $G_{c\ell,\boldsymbol{u}}$ with a spreading kernel φ [5, 6], $V(\boldsymbol{k}) \approx \sum_{\boldsymbol{u}} \varphi(\boldsymbol{k}-\boldsymbol{u})G_{c\ell,\boldsymbol{u}}$ (after pre-compensation $1/\hat{\varphi}$), and

Proposition 1 (derivatives are exact on the interpolant)**.** $\partial^\alpha V(\boldsymbol{k}) \approx \sum_{\boldsymbol{u}} \partial^\alpha \varphi(\boldsymbol{k}-\boldsymbol{u})\, G_{c\ell,\boldsymbol{u}}$ *is the exact derivative of the interpolant, so Equation* (10) *turns into a single gather with a* per-sample kernel

$$\begin{aligned} \eta_{j,\boldsymbol{u}} &= \varphi(\boldsymbol{k}'_j - \boldsymbol{u}) + \mathrm{i} \sum_{pq} \Delta C_{j,pq}\, \partial_p \partial_q \varphi(\boldsymbol{k}'_j - \boldsymbol{u}) \\ &\quad - \tfrac{1}{2} \sum_{pqrs} \Delta C_{j,pq} \Delta C_{j,rs}\, \partial_{pqrs} \varphi(\boldsymbol{k}'_j - \boldsymbol{u}), \\ (Ax)_{cj} &= \mathrm{e}^{-\mathrm{i}\phi_{0j}} \sum_{\boldsymbol{u}} \eta_{j,\boldsymbol{u}} \sum_{\ell} a_\ell(\tau_j) G_{c\ell,\boldsymbol{u}}. \end{aligned} \tag{11}$$

The kernel $\eta_{j,\boldsymbol{u}}$ does not depend on the coil or the segment: it is evaluated *once per (sample, grid point)* and reused for all $C \cdot L$ grids. It is formed in registers from the 6 numbers ΔC_j: the first-order term involves the 6 distinct second derivatives, the second-order term the 15 distinct fourth derivatives (monomials of degree 4 in three variables).

Adjoint. A^{H} is the exact transpose: scatter $\overline{\eta_{j,\boldsymbol{u}}}\, \overline{a_\ell(\tau_j)}\, \mathrm{e}^{+\mathrm{i}\phi_{0j}}\, y_{cj}$ into the L grids, one batched inverse FFT, crop, multiply by $\overline{D_\ell}$, sum over ℓ, by $\overline{S_c}$, sum over c (a fused multi-input crop pass). A^{H} is consistent with A to f32 rounding for all temporal options below.

Spreading kernel. The derivative form needs $\partial^\alpha \varphi$ up to order 4 bounded. The exponential-of-semicircle kernel $\mathrm{e}^{\beta(\sqrt{1-z^2}-1)}$ is C^0 with $\sqrt{\cdot}$ singular derivatives at $|z|=1$; we clamp its derivative weights to zero for $|z| > 0.98$ (without the clamp the 4th-derivative error is 46× larger, 10^{-3} interpolation error). The Kaiser–Bessel kernel $I_0(\beta\sqrt{1-z^2})$ is entire in z and needs no clamp. The accuracy of the *interpolated* derivative degrades with order and with oversampling [measured, 1-D, relative error of value / ∂^2]:

oversampling σ (grid volume vs. σ=2)	width W	value	∂^2
2.0 (1.00)	7	$9{\cdot}10^{-7}$	$8{\cdot}10^{-5}$
2.0	9	$1{\cdot}10^{-8}$	$1.4{\cdot}10^{-6}$
1.5 (0.42)	10	$6{\cdot}10^{-8}$	$6{\cdot}10^{-5}$
1.25 (0.24)	10	$2{\cdot}10^{-6}$	$3{\cdot}10^{-5}$

Since the correction term itself is $\lesssim 0.2$ rad, ∂^2 accuracy 10^{-3} suffices, so the operator runs at $\sigma = 1.25$, $W = 10$ on a grid with a quarter of the $\sigma = 2$ volume.

Temporal interpolation. A sample acquired at cardiac time t between frames: $x(\boldsymbol{r}, t) = \sum_f B_f(t)\, x_f(\boldsymbol{r})$ (periodic linear or cubic B-spline in cardiac phase, nodes at the frames). The sample then gathers from the $N_f = 2$ (linear) or 4 (cubic) neighbouring frame grids, with coefficients $a_\ell(\tau_j) B_f(t_j)$ over $L \times N_f$ channels. FFTs scale with the number of frames, not with N_f; gather/scatter work scales with N_f. This removes the encode-train skew (encodes of one frame are not simultaneous) and the bin-width blur from the model; velocities are formed from $x_e(t)$ and $x_{\mathrm{ref}}(t)$ at the same t.

Implementation facts relevant to cost. (i) Fine grid stored channel-interleaved, [point][$c \cdot L + \ell$], one partial-axis FFT over all channels, pad/crop fused over channels. (ii) Exact tile→sample lists (CSR): every 4^3 tile of the grid lists the samples whose W^3 support meets it; long lists are split into work items. (iii) Per sample one 48-byte record (shifted $\boldsymbol{k}'$, τ, ϕ_0, ΔC); the 15 second-order

coefficients are formed in the kernel. (iv) Adjoint: thread per (tile point, coil) with L accumulators, no atomics. (v) Launch shapes come from a planner bounded by 48 KB shared memory per block and are refined by a short autotune; nothing is tuned to one GPU generation (kernels are generated at run time).

Accuracy and cost [measured]. Accuracy against an f64 direct sum (16×14×10 voxels, 3 coils, B0 + τ-linear quadratic + per-sample $\Delta C + \phi_0$):

Taylor order of ΔC term	$L = 6$	$L = 8$	$L = 10$
0 (no correction)	$3.8{\cdot}10^{-2}$	–	–
1	$2.4{\cdot}10^{-3}$	$1.7{\cdot}10^{-3}$	$1.7{\cdot}10^{-3}$
2	$1.8{\cdot}10^{-3}$	$8.5{\cdot}10^{-5}$	$7.7{\cdot}10^{-5}$

(order 0 with $L = 4$: $7.2{\cdot}10^{-2}$.) The floor of each order is the dropped next term of Equation (10), independent of L; L controls only the τ-dependent part. With temporal interpolation (L=10, order 2): static $7.9{\cdot}10^{-5}$, linear (5 frames) $7.4{\cdot}10^{-5}$, cubic (6 frames) $7.8{\cdot}10^{-5}$. Cost on an RTX 4090, 180×180×28 grid, $M = 2.2{\cdot}10^5$ samples, 12 coils (ms per call):

operator	forward	adjoint
plain NUFFT, σ=2, W=6	6.0	24.7
plain NUFFT, σ=1.25, W=10	7.1	89
segmented, L=1, order 0	5.7	8.7
segmented, L=8, orders 0 / 1 / 2	30.1 / 30.9 / 31.7	31.2 / 31.8 / 30.1
segmented, L=10 / 12, order 2	40.4 / 49.2	35.3 / 41.0

The second-order correction is essentially free ($o2 \approx o0$) because the weights are shared by all coils and segments; the cost is the L FFT pairs and the memory-bound pad/FFT/crop passes (about half of the call). Time-resolved: 20 frames, $4.4{\cdot}10^6$ samples per encode, $L = 8$, order 2: 0.98/0.93 s (linear) and 1.9/1.9 s (cubic) per forward/adjoint pair, against 0.62/0.59 s for 20 static per-frame calls.

1.4 Algebra of the normal operator: Toeplitz structure, commutators, Weyl symbol

The reconstruction needs $\mathsf{N} = A^{\mathsf{H}}A$ in two roles: the iteration gradient $\nabla f(x) = \mathsf{N}x - A^{\mathsf{H}}y$ and the preconditioner. Because A_j is coil-free and the coil maps are pointwise multiplications, with $\mathsf{N}_0 = A_{\text{samp}}^{\mathsf{H}} A_{\text{samp}}$ the sample-side Gram operator,

$$\mathsf{N} = \sum_c \overline{S_c}\, \mathsf{N}_0\, S_c, \tag{12}$$

i.e. the coil sum is the *outermost* loop; the segment interpolators, kernels, and everything below are coil-free. Hence everything in this subsection is stated for one sample-side operator $\mathsf{N}_0 =: \mathsf{N}$ with sample weights $\varpi_j \geq 0$ ($\varpi_j \equiv 1$ in our pipeline; the choice of ϖ is a preconditioning question, not part of the model).

Setting

Let X_a be multiplication by coordinate r_a, $\mathsf{P} = -\mathrm{i}\nabla$ the translation generator, and for a kernel $g(\boldsymbol{\Delta})$ let $\mathsf{T}[g]\rho(\boldsymbol{r}) = \sum_{\boldsymbol{r}'} g(\boldsymbol{r}-\boldsymbol{r}')\rho(\boldsymbol{r}')$ be the Toeplitz (convolution) operator. For a plane wave $\mathsf{T}[\mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}]\rho = \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{r}}\hat{\rho}(\boldsymbol{k}_j)$ is rank one. From Equation (1), with $\Phi_j = \Theta_j - \boldsymbol{k}_j\cdot\boldsymbol{r}$ the non-trajectory part,

$$\mathsf{N}(\boldsymbol{r},\boldsymbol{r}') = \sum_j \varpi_j\, \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot(\boldsymbol{r}-\boldsymbol{r}')}\, \mathrm{e}^{\mathrm{i}(\Phi_j(\boldsymbol{r})-\Phi_j(\boldsymbol{r}'))}, \qquad \mathsf{N} = \sum_j \varpi_j\, \mathsf{Q}_j\, \mathsf{T}[\mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}]\, \mathsf{Q}_j^{\mathsf{H}}, \quad \mathsf{Q}_j = \mathrm{e}^{+\mathrm{i}\Phi_j(\mathsf{X})}. \tag{13}$$

For $\Phi_j \equiv 0$ this is $\mathsf{T}[\sum_j \varpi_j \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}]$: the PSF kernel, a single FFT-diagonal operator (the "Toeplitz embedding"). All physics enters through the conjugations Q_j. For the exact statements below we take $\Phi_j = \sum_q w_{jq} m_q$, the quadratic (concomitant) part; the $\omega(\boldsymbol{r})\tau_j$ part is carried by the segment diagonals (Section 1.2) or treated at first order (remark in Section 1.4).

Lemma 2 (commutator of a coordinate with a Toeplitz operator)**.** $[\mathsf{X}_a, \mathsf{T}[g]] = \mathsf{T}[\Delta_a g]$, *hence* $\mathrm{ad}_{\mathsf{X}}^{\beta}\mathsf{T}[g] = \mathsf{T}[\boldsymbol{\Delta}^{\beta} g]$ *for any multi-index* β.

Proof. $([\mathsf{X}_a, \mathsf{T}[g]]\rho)(\boldsymbol{r}) = \sum_{\boldsymbol{r}'}(r_a - r'_a) g(\boldsymbol{r}-\boldsymbol{r}')\rho(\boldsymbol{r}')$. Iterate. (Checked numerically to $6{\cdot}10^{-14}$.) □

Lemma 3 (midpoint identity for quadratic phases)**.** *If* $\Phi(\boldsymbol{r}) = \frac{1}{2}\boldsymbol{r}^{\top} M \boldsymbol{r} + \boldsymbol{b}\cdot\boldsymbol{r} + c$ *with* $M = M^{\top}$, *then with* $\bar{\boldsymbol{r}} = (\boldsymbol{r}+\boldsymbol{r}')/2$ *and* $\boldsymbol{\Delta} = \boldsymbol{r} - \boldsymbol{r}'$,

$$\Phi(\boldsymbol{r}) - \Phi(\boldsymbol{r}') = \nabla\Phi(\bar{\boldsymbol{r}})\cdot\boldsymbol{\Delta} = (M\bar{\boldsymbol{r}} + \boldsymbol{b})\cdot\boldsymbol{\Delta} \qquad \textit{(exact).}$$

Proof. $\frac{1}{2}(\boldsymbol{r}^{\top} M \boldsymbol{r} - \boldsymbol{r}'^{\top} M \boldsymbol{r}') = \frac{1}{2}(\boldsymbol{r}+\boldsymbol{r}')^{\top} M (\boldsymbol{r}-\boldsymbol{r}')$ for symmetric M; the linear part is $\boldsymbol{b}\cdot\boldsymbol{\Delta}$. □

The concomitant monomials m_q are quadratic, hence Lemma 3 applies to $\Phi_j = \sum_q w_{jq} m_q$ with $M_j = \sum_q w_{jq} H_q$, $H_q = \nabla^2 m_q$ constant (e.g. $H_{z^2} = \mathrm{diag}(0,0,2)$). The same holds for the quadratic part of any smooth ψ after Taylor expansion about the FOV centre; for general ψ the identity holds to first order with $\nabla\psi(\bar{\boldsymbol{r}})$ and remainder $O(\Delta^3\partial^3\psi)$.

Proposition 4 (exact Weyl symbol of N)**.** *Define the Weyl symbol of an operator with kernel* K *by* $\sigma_K(\bar{\boldsymbol{r}},\boldsymbol{k}) = \sum_{\boldsymbol{\Delta}} K(\bar{\boldsymbol{r}} + \boldsymbol{\Delta}/2, \bar{\boldsymbol{r}} - \boldsymbol{\Delta}/2)\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\Delta}}$. *For quadratic* Φ_j,

$$\sigma_{\mathsf{N}}(\bar{\boldsymbol{r}},\boldsymbol{k}) = \sum_j \varpi_j\, \delta\big(\boldsymbol{k} - \boldsymbol{k}_j - M_j\bar{\boldsymbol{r}} - \boldsymbol{b}_j\big), \qquad \mathsf{N} = \sum_j \varpi_j\, \delta(\Pi_j - \boldsymbol{k}_j), \quad \Pi_j = \mathsf{P} - M_j\mathsf{X} - \boldsymbol{b}_j, \tag{14}$$

with no $\hbar$*-type corrections. The three components of* Π_j *commute because* M_j *is symmetric.*

Proof. By Lemma 3 the kernel of the j-th term is $\varpi_j \mathrm{e}^{\mathrm{i}(\boldsymbol{k}_j + M_j\bar{\boldsymbol{r}} + \boldsymbol{b}_j)\cdot\boldsymbol{\Delta}}$, whose symbol is the delta function in Equation (14). For the operator form, $\mathsf{Q}_j\mathsf{P}\mathsf{Q}_j^{\mathsf{H}} = \mathsf{P} - \nabla\Phi_j(\mathsf{X})$ (compute $\mathrm{e}^{\mathrm{i}\Phi}(-\mathrm{i}\nabla)(\mathrm{e}^{-\mathrm{i}\Phi} f) = \mathsf{P}f - \nabla\Phi\, f$), so $\Pi_j = \mathsf{Q}_j\mathsf{P}\mathsf{Q}_j^{\mathsf{H}}$ with $\nabla\Phi_j(\mathsf{X}) = M_j\mathsf{X} + \boldsymbol{b}_j$, and $\mathsf{Q}_j\mathsf{T}[\mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}]\mathsf{Q}_j^{\mathsf{H}}$ is the rank-one projector on the eigenfunction $\mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{r} + \mathrm{i}\Phi_j(\boldsymbol{r})}$ of Π_j with eigenvalue $\boldsymbol{k}_j$. Commutation: $[\Pi_{j,a}, \Pi_{j,b}] = [P_a, -(M_j\mathsf{X})_b] + [-(M_j\mathsf{X})_a, P_b] = \mathrm{i}(M_{j,ba} - M_{j,ab}) = 0$. □

Interpretation. Conjugation by $\mathrm{e}^{\mathrm{i}\Phi(\mathsf{X})}$ with quadratic Φ is a metaplectic operator, which acts on phase space as the *linear symplectic shear* $(\boldsymbol{r},\boldsymbol{k}) \mapsto (\boldsymbol{r}, \boldsymbol{k} - \nabla\Phi(\boldsymbol{r}))$; Weyl symbols transform exactly under linear symplectic maps (Egorov's theorem is exact for metaplectic operators [8]). So the concomitant field turns the sample $\boldsymbol{k}_j$ into a *plane in phase space that is tilted by* M_j: at position $\bar{\boldsymbol{r}}$ the sample is seen at the shifted frequency $\boldsymbol{k}_j + M_j\bar{\boldsymbol{r}} + \boldsymbol{b}_j$. This is the exact content of "the PSF is spatially varying: the effective trajectory at $\bar{\boldsymbol{r}}$ is $\boldsymbol{k}_j^{\mathrm{eff}}(\bar{\boldsymbol{r}}) = \boldsymbol{k}_j + M_j\bar{\boldsymbol{r}} + \boldsymbol{b}_j$", linear in $\bar{\boldsymbol{r}}$. For non-polynomial ψ (B0, GNL) the shear is nonlinear and composition produces $\hbar^2$-type terms of order $\partial^3\psi$: this is the origin of the remainder in Lemma 3.

Proposition 5 (commutator expansion)**.** *Let* $\Phi_j = \sum_q w_{jq} m_q$, $\alpha \in \mathbb{N}^4$ *a multi-index,* $\alpha! = \prod \alpha_q!$, $w_j^\alpha = \prod w_{jq}^{\alpha_q}$, *and define the moment-weighted Toeplitz operators* $\mathsf{T}^\alpha = \mathsf{T}[g^\alpha]$, $g^\alpha(\boldsymbol{\Delta}) = \sum_j \varpi_j w_j^\alpha \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}$. *Then*

$$\mathsf{N} \;=\; \sum_\alpha \frac{\mathrm{i}^{|\alpha|}}{\alpha!}\, \mathrm{ad}_{m_1}^{\alpha_1}\cdots \mathrm{ad}_{m_4}^{\alpha_4}\, \mathsf{T}^\alpha \;=\; \mathsf{T}^0 + \mathrm{i}\sum_q [m_q(\mathsf{X}), \mathsf{T}^{e_q}] - \tfrac{1}{2}\sum_{q,q'} [m_q, [m_{q'}, \mathsf{T}^{e_q+e_{q'}}]] + \cdots \tag{15}$$

where $\mathrm{ad}_m = [m(\mathsf{X}), \cdot\,]$.

Proof. Hadamard's lemma $\mathrm{e}^{\mathrm{i}A} T \mathrm{e}^{-\mathrm{i}A} = \sum_n \frac{\mathrm{i}^n}{n!} \mathrm{ad}_A^n T$ with $A = \Phi_j(\mathsf{X}) = \sum_q w_{jq} m_q(\mathsf{X})$. The $m_q(\mathsf{X})$ are commuting multiplication operators, so ad_{m_q} and $\mathrm{ad}_{m_{q'}}$ commute (Jacobi: $[\mathrm{ad}_A, \mathrm{ad}_B] = \mathrm{ad}_{[A,B]} = 0$), and the multinomial theorem gives $\mathrm{ad}_A^n = \sum_{|\alpha|=n} \frac{n!}{\alpha!} w_j^\alpha \mathrm{ad}_m^\alpha$. Sum over j. (Checked: the series reproduces the exact dense N to $5\cdot 10^{-12}$ at order 8 on a 1-D test with $\max \Phi = 0.4$.) □

Each term of Equation (15) is a nested commutator of *diagonal* operators (the m_q, products of coordinates) with *base* Toeplitz operators whose kernels g^α are the sample PSF re-weighted by monomials of the coefficients. Order exactly n has $\binom{n+3}{3}$ multi-indices over the four monomials, so $\binom{n+4}{4}$ base kernels through order n – not L^2.

The Δ-form: diagonals are coordinate monomials only. By Lemma 3 the pair phase vanishes on the diagonal $\boldsymbol{\Delta} = 0$ and is *linear in* $\boldsymbol{\Delta}$; expand $\mathrm{e}^{\mathrm{i}(\Phi_j(\boldsymbol{r}) - \Phi_j(\boldsymbol{r}'))}$ in the pair phase instead of in Φ_j:

$$\mathsf{N} = \sum_{n\ge 0} \mathsf{K}_n, \qquad K_n(\boldsymbol{r}, \boldsymbol{r}') = \frac{\mathrm{i}^n}{n!} \sum_j \varpi_j \big[(M_j \bar{\boldsymbol{r}} + \boldsymbol{b}_j)\cdot\boldsymbol{\Delta}\big]^n \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}} = \sum_{|\beta|\le n} \bar{\boldsymbol{r}}^\beta\, g_{n,\beta}(\boldsymbol{\Delta}), \tag{16}$$

with $g_{n,\beta}$ *moment kernels* (the sample PSF weighted by polynomials in $\boldsymbol{\Delta}$ and in the coefficients of M_j, $\boldsymbol{b}_j$). Since $\bar{\boldsymbol{r}}^\beta = 2^{-|\beta|} \sum_{\gamma\le\beta} \binom{\beta}{\gamma} r^\gamma r'^{\beta-\gamma}$, the operator with kernel $\bar{\boldsymbol{r}}^\beta g(\boldsymbol{\Delta})$ is the *Weyl-symmetrised product* $2^{-|\beta|} \sum_{\gamma\le\beta} \binom{\beta}{\gamma} \mathsf{X}^\gamma\, \mathsf{T}[g]\, \mathsf{X}^{\beta-\gamma}$; for $|\beta| = 1$ it is $\frac{1}{2}\{\mathsf{X}_a, \mathsf{T}[g]\}$. Consequences:

1. *Exact Hermitian structure.* Under $\boldsymbol{r} \leftrightarrow \boldsymbol{r}'$ ($\bar{\boldsymbol{r}}$ fixed, $\boldsymbol{\Delta} \to -\boldsymbol{\Delta}$) $K_n(\boldsymbol{r}', \boldsymbol{r}) = \overline{K_n(\boldsymbol{r}, \boldsymbol{r}')}$ for every n, so *every truncation is exactly Hermitian*, with no symmetrisation step. (Checked, 0 to machine precision.)

2. *FFT count.* Order n needs the transforms of $\mathsf{X}^\beta \rho$ for $|\beta| \le n$ and one inverse transform per output monomial: $\binom{n+3}{3}$ FFT pairs (4, 10, 20, 35 for $n = 1, \ldots, 4$), independent of the number of samples and of L.

3. *Truncation.* $|\Phi_j(\boldsymbol{r}) - \Phi_j(\boldsymbol{r}')| \le 2\Phi_{\max}$ gives the pointwise remainder $\le \sum_j \varpi_j (2\Phi_{\max})^{n+1}/(n+1)!$ – the expansion parameter is the *pair* phase, which is small where N has weight (near the diagonal), rather than Φ itself.

First order, explicitly. With $\Phi_j = \sum_q w_{jq} m_q$ and Lemma 3,

$$\mathrm{i}\sum_q [m_q(\mathsf{X}), \mathsf{T}^{e_q}] = \frac{\mathrm{i}}{2} \sum_q \sum_{a,b} (H_q)_{ab} \big\{\mathsf{X}_a,\ \mathsf{T}[\Delta_b\, g^{e_q}]\big\} = \frac{\mathrm{i}}{2} \sum_{q,a} \big\{\partial_a m_q(\mathsf{X}),\ \mathsf{T}[\Delta_a g^{e_q}]\big\}$$

(using $[m_q, \mathsf{T}[g]]$ has kernel $\nabla m_q(\bar{\boldsymbol{r}}) \cdot \boldsymbol{\Delta}\, g$ by Lemma 3, and ∇m_q is linear). The distinct kernels are $\Delta_z g^1; \Delta_x g^2, \Delta_y g^2; \Delta_x g^3, \Delta_z g^3; \Delta_y g^4, \Delta_z g^4$: **7 kernels**. The evaluation is: forward transforms

of ρ, $x\rho$, $y\rho$, $z\rho$ (4); in k-space multiply by the 7 kernels (and by g^0 for the base term); inverse transforms: one for the sum of everything that is already multiplied from the right by a coordinate, and one per coordinate a for the terms that still need the left multiplication X_a (4 inverse). **4 FFT pairs** per coil recover $\approx 94\%$ of the concomitant effect at $\Phi_{\max} = 0.3$ rad; 10 pairs (order 2) $\approx 99.7\%$ [measured, 8^3 toy].

Fourier-domain reading. Under $\mathsf{X}_a \mapsto \mathrm{i}\partial_{k_a}$ the order-1 term is a first-order differential operator in $\boldsymbol{k}$ acting on $\hat{\rho}$, $-\mathrm{i}\sum_{q,a,b}(H_q)_{ab}\big[\partial_b \hat{g}^q\, \partial_a \hat{\rho} + \frac{1}{2}\partial_a\partial_b \hat{g}^q\, \hat{\rho}\big]$ (from $\mathsf{X}_a \mapsto \mathrm{i}\partial_{k_a}$, $\mathsf{T}[\Delta_a g] \mapsto \mathrm{i}\partial_a \hat{g}$) with $\hat{g}^q$ the gridded sample density weighted by w_q. The kernel derivatives are exact (multiply the kernel by Δ_b before its FFT); *finite differences of $\hat{\rho}$ on the FFT grid are a trap*: the image occupies up to half of the $2N$ padded box, so $\hat{\rho}$ is sampled at only $2\times$ Nyquist and a central difference near the FOV edge has $\sim 40\%$ error, while $\mathcal{F}\{x_a\rho\}$ is exact and cheaper. The multiplier $\hat{g}$ is rough (it fills the whole $2N$ period) and cannot be differenced either.

How the pieces relate

- *Sample-domain vs. normal-operator corrections are the same Taylor idea in different variables.* On the A side there is only the *single* phase $\boldsymbol{r}^\top \Delta C \boldsymbol{r}$ (expansion centre: the FOV centre), so the series Equation (10) is only usable for the small residual; the large smooth part is carried exactly by the segments. On the N side the natural variable is the *pair* phase (Lemma 3), which vanishes on the diagonal; the expansion is correspondingly twice as efficient at equal FFT count [measured, below]. Both have the same structure: second-derivative weights in $\boldsymbol{k}$ $\leftrightarrow$ coordinate-monomial diagonals in $\boldsymbol{r}$.

- *Off-resonance at first order.* For $\psi = \omega(\boldsymbol{r})\tau_j$: $\mathsf{N} \approx \mathsf{N}^0 + \frac{\mathrm{i}}{2}\sum_a \{\partial_a\omega(\mathsf{X}), \mathsf{T}[\Delta_a g^\tau]\}$ with $g^\tau = \sum_j \varpi_j \tau_j \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}$: 3 diagonals, 3 kernels; remainder $O(\Delta^3\partial^3\omega)$. Useful as a cheap correction, not as a replacement for segments at $\omega\tau$ of several radians.

- *Time-segmented normal operator.* Using the segment form, $\mathsf{N} \approx \sum_{\ell,\ell'} D_\ell^{\mathsf{H}}\, \mathsf{T}[g_{\ell\ell'}]\, D_{\ell'}$ with $g_{\ell\ell'} = \sum_j \varpi_j \overline{a_\ell(\tau_j)} a_{\ell'}(\tau_j) \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}$ (Hermitian in $\ell \leftrightarrow \ell'$: $L(L+1)/2$ kernels, L FFT pairs per coil). This is the Fessler–Sutton–Noll form [4]. For the *time-resolved* reconstruction (here ≈ 80 sets = 20 frames $\times$ 4 encodes) it needs $\mathcal{O}(L^2)$ kernels *per set*, which does not fit GPU memory, while each set has few samples ($\approx 2{\cdot}10^5$ versus $9{\cdot}10^5$ voxels), so the gather/scatter the Toeplitz form saves is cheap. Both forms need the same L FFT pairs per coil. **For time-resolved reconstruction we therefore use the NUFFT form (Section 1.3)**; the Toeplitz form is attractive for large-K, few-set problems (time-averaged warm start).

- *Taylor of $\mathrm{e}^{-\mathrm{i}\Phi}$ ("multinomial") and its Hankel structure.* Expanding each factor, $\mathrm{e}^{-\mathrm{i}\Phi_j} = \sum_\alpha c_\alpha(j) F_\alpha$, gives $\mathsf{N} = \sum_{\alpha,\alpha'} \frac{\mathrm{i}^{|\alpha|-|\alpha'|}}{\alpha!\alpha'!} F_\alpha^{\mathsf{H}}\, \mathsf{T}^{\alpha+\alpha'} F_{\alpha'}$, with the kernel depending only on $\alpha+\alpha'$ (Hankel): $\binom{4+2P}{2P}$ kernels instead of $L(L+1)/2$, but $\binom{4+P}{P}$ fields (FFT pairs) at order P – twice as many pairs as the Weyl form at equal accuracy.

Operator accuracy versus FFT pairs [measured; 8^3 voxels, 960 samples, random $SO(3)$-rotated spirals, Φ rescaled; relative error $\|\mathsf{N} - \mathsf{N}_{\text{approx}}\|_F / \|\mathsf{N} - \mathsf{N}^0\|_F$, 1 = ignoring the concomitant field].

method (FFT pairs)	$\Phi_{\max}$=0.3	1	3
Taylor of $e^{-i\Phi}$, P=2 (15)	$6.2{\cdot}10^{-3}$	$7.6{\cdot}10^{-2}$	1.1
Taylor, P=3 (35)	$4.5{\cdot}10^{-4}$	$1.6{\cdot}10^{-2}$	0.37
Weyl Δ-form, n=1 (4)	$6.2{\cdot}10^{-2}$	0.21	0.61
Weyl, n=2 (10)	$3.5{\cdot}10^{-3}$	$3.8{\cdot}10^{-2}$	0.34
Weyl, n=3 (20)	$1.7{\cdot}10^{-4}$	$6.3{\cdot}10^{-3}$	0.17
Weyl, n=4 (35)	$7.2{\cdot}10^{-6}$	$8.9{\cdot}10^{-4}$	$7.3{\cdot}10^{-2}$
SVD–Tucker L=6 (6)	$4.4{\cdot}10^{-3}$	$1.5{\cdot}10^{-2}$	$4.6{\cdot}10^{-2}$
SVD–Tucker L=12 (12)	$5.9{\cdot}10^{-4}$	$2.1{\cdot}10^{-3}$	$9.8{\cdot}10^{-3}$

The concomitant correction itself is small relative to N in this test ($\|\mathsf{N}-\mathsf{N}^0\|/\|\mathsf{N}\| = 0.8$, 2.8, 7.9%). The Weyl form needs about half the FFT pairs of the Taylor form at equal error; the SVD (Tucker) basis needs the fewest pairs but a data-dependent basis and $L(L+1)/2$ kernels. One-sided product-convolution expansions [7] (a single diagonal, $\mathsf{N} \approx \sum_\ell D_\ell \mathsf{T}[h_\ell]$) were measured at 2–5× more terms than the two-sided Tucker form and are not used. The order needed grows roughly linearly in $\Phi_{\max}$ (consistent with the pair-phase bound $2\Phi_{\max}$). Honest limits: no method found has an FFT-pair count independent of the phase range – the Slepian number of (visited coefficient set × range of the pair phase) $\approx (\Phi_{\max}/\pi) \cdot c$ sets a floor of $\sim L$; subdividing the sample set (butterfly-type splitting by time or direction) only increases the total number of bases (42 → 51/80/132 for 2/4/8 time bins at $\Phi \approx 9$ rad).

Stencil (chirp-aware) alternative for large Φ. The same exactness can be read in k-space: for quadratic phases the Gram operator on the oversampled grid is a *banded* matrix $\hat{\mathsf{N}}(\boldsymbol{k}, \boldsymbol{k}+\boldsymbol{\delta}) = \nu_{\boldsymbol{\delta}}(\boldsymbol{k})$, $|\boldsymbol{\delta}| \lesssim s$ with s the chirp's spectral radius, and $\nu_{\boldsymbol{\delta}} = \sum_\alpha c_{\boldsymbol{\delta}\alpha} \hat{g}^\alpha$ with a universal table $c_{\boldsymbol{\delta}\alpha}$ (independent of the trajectory). The NUFFT counterpart absorbs the chirp $e^{-i\Phi_j}$ into a per-sample stencil $\Gamma_j = \varphi \star \hat{g}(\cdot; \boldsymbol{w}_j)$ with a single FFT pair per coil, independent of L and of Φ; its width is set by the chirp spectrum (taps per axis for 10^{-3}: 8, 10, 13–14, 18 at $\Phi_{\max} = 1, 2, 4, 8$ rad, compared with 4 for a plain kernel at $\sigma = 2$), and the stencil spectrum must be fitted by least squares on the FOV (a DFT of a windowed chirp needs $\sim 3\times$ more taps). It is the only route with zero stored kernels and an FFT count independent of Φ, and a competitive choice for strongly undersampled sets at $\Phi \gtrsim 2$ rad; in the production pipeline the segmented operator above replaces it because the readout-dependent remainder after the split Equation (5) is small (second-order derivative weights suffice).

1.5 Using the operators inside the preconditioned plug-and-play iteration

We solve $\min_x \frac{1}{2}\|Ax - y\|^2 + \lambda R(x)$ by the preconditioned plug-and-play iteration (P^{2}nP [9])

$$x_{k+1} = \mathcal{D}\big(x_k - \alpha\, \mathsf{P}_k\, g_k\big), \qquad g_k = \nabla f(x_k) = \mathsf{N} x_k - A^{\mathsf{H}} y, \tag{17}$$

with $\mathcal{D}$ a learned or classical denoiser and P_k a preconditioner built from secant pairs $(s_k, u_k) = (x_k - x_{k-1},\, g_k - g_{k-1})$ (ZMSHR1, a rank-one update of a base metric; one operator application per iteration). There is no density compensation: for a spiral sampling every readout dwells at $\boldsymbol{k} = 0$, so N has a dynamic range of 10^4–10^6 and the *preconditioner* replaces the weights.

Base metric. A circulant (T. Chan) approximation of N using the exact Fourier diagonal $\langle e_{\boldsymbol{k}}, \mathsf{N} e_{\boldsymbol{k}} \rangle = (\rho \circledast P_{\text{coil}})(\boldsymbol{k})$, the sample density convolved with the coil-map power spectrum. Three details are essential: (i) the $\boldsymbol{k} = 0$ density spike is smeared onto neighbouring modes by the maps, so use

the convolved diagonal; (ii) raw Biot–Savart maps blow up near the loops, so use ESPIRiT-type normalised maps $\sum_c |S_c|^2 = 1$ on the dilated support; (iii) floor the unsampled corners. These reduced $\lambda_{\max}(M\mathsf{N})$ from 38 to 9.

Lagged correction (control variate). The data-consistency gradient is the expensive operation. Write N_ℓ for a cheap surrogate (plain Toeplitz, a low-order Weyl form, or $L = 1$) and keep an *anchor* x_a at which the exact operator was applied:

$$g_k \;\approx\; \mathsf{N}_\ell\,(x_k - x_a) + g_{\text{exact}}(x_a), \qquad \text{refresh } x_a \leftarrow x_k \;\text{ when }\; \|x_k - x_a\| > \tau\|x_k\|. \tag{18}$$

This removes the surrogate bias at the fixed point and only the Lipschitz mismatch times $\|x_k - x_a\|$ remains. It matters because $\mathsf{N} - \mathsf{N}^0$ is a small perturbation (cf. the table above) but *cheap-only gradients are unsafe with a quasi-Newton preconditioner*: secant pairs learned from a mismatched operator are applied along directions where N has small eigenvalues, so errors that are tiny on typical images ($3 \cdot 10^{-4}$ for order-2 Weyl) still blow up. Measured on the 16^3 toy problem at $\Phi_{\max} = 3$ rad, 100 iterations, no denoiser (cost in units of one plain-Toeplitz application per coil, exact = 39 units):

gradient strategy	NRMSE	exact applications	cost
exact operator every iteration	0.213	100	15600
plain surrogate only	1.29	0	400
Weyl order 2 only	$7.7{\cdot}10^4$	0	12800
lagged plain, τ=0.1	0.218	12	2320
lagged plain, τ=0.05	0.214	19	3440

The lagged plain surrogate is 4.5–6.7× cheaper for the same error; higher-order surrogates do not pay at the measured cost ratio. This is a statement about the data-consistency and preconditioner machinery on a toy problem with identity denoiser; with the real denoiser the denoiser's bias dominates the plateau, and the study must be repeated at $N \geq 64$ with real coil maps. In the full pipeline the exact operator is `SegNufft`, and the surrogate a reduced-L `SegNufft` or a plain Toeplitz form; this combination is a design, the measurements above use the toy operators.

1.6 Velocity parametrisation on top of the operator

The operator maps a *magnetisation image* $x_e(\boldsymbol{r}, \tau)$ per encode e to data. Instead of E complex images ($E = 4$, 7, 13) the unknowns can be the physical quantities,

$$x_e(\boldsymbol{r}, \tau) = z(\boldsymbol{r}, \tau)\, \mathrm{e}^{\mathrm{i}\varphi_e(\boldsymbol{r})}\, \mathrm{e}^{-\mathrm{i}\gamma\, M_{1,e}\cdot v(\boldsymbol{r},\tau)}, \tag{19}$$

with $z \in \mathbb{C}$ the reference signal (magnitude and every phase shared by all encodes), $v \in \mathbb{R}^3$ the velocity, and $\varphi_e = \sum_p a_{e,p} P_p(\boldsymbol{r})$ a low-order Legendre residual phase per encode (eddy currents, imperfect Maxwell/B0/GNL correction); z, v in periodic B-spline bases in cardiac time ($K_z \leq K_v$ nodes, independent of the operator's frames). The model is *linear in* z for fixed (v, a), so no positivity or phase-wrap constraint enters; it has 5 real unknowns per voxel and node instead of $2E$; encodes are coupled through one v, so dual-VENC unwrapping is part of the solve rather than a per-voxel rounding afterwards. The first moment of the *readout itself* ($M_1^{\text{ro}}(\tau)\cdot v$, up to ~ 1 cm of displacement at 1 m/s over 10 ms) is a known function of the current velocity iterate and is applied as a lagged per-segment warp inside D_ℓ (same segments, no extra FFT). The data term is evaluated through the operator of Section 1.3; the Gauss–Newton Hessian-vector product and a pointwise

Gauss–Newton metric feed the same ZMSHR1 machinery. GNL is handled by reconstructing in apparent coordinates $\boldsymbol{r} = u(\boldsymbol{r}_{\text{true}})$, warping back, and correcting the velocity by the Jacobian J^{-1} (the known GNL VENC error).

1.7 Validation summary [measured on simulated data with a model-independent truth simulator]

The truth data are produced by a brute-force GPU direct sum over a fine point cloud (no NUFFT, no Toeplitz, nothing shared with the reconstruction operators; checked against an f64 CPU sum to $1.2{\cdot}10^{-5}$) that includes the exact gradient moments (M_0, M_1, M_2), B0 from susceptibility, GNL, and the concomitant phase of the whole waveform; the reconstruction therefore has genuine model mismatch (static-within-readout motion, intravoxel dephasing, estimated maps).

- *Physics ablation, time-averaged least squares per encode* (subject R4, 60 s scan, 100 PCGLS iterations, velocity RMSE in the lumen): no physics 0.211 m/s; B0 only 0.249 (slower convergence); B0 + concomitant 0.131; B0 + concomitant on data without B0/GNL 0.060. The ~ 0.1 m/s offsets removed by the concomitant model are the Maxwell velocity offsets of the bipolars.
- *Operator validation on a CFD-based case* (RACLETTE case n001, dense model-exact data with B0 + concomitant + GNL): full model image error $4.3 \cdot 10^{-3}$ (CG-limited), velocity RMSE 2.4 mm/s, bias 0.0000.
- *Velocity model* (R4, 7-point dual-VENC 1.0/3.5 m/s, 60 s, $R \approx 8$ per frame, 30 iterations, *no prior*): 0.138 m/s with 8 cubic nodes for z and v and a VENC continuation, against 0.183 m/s for the complex model with a locally-low-rank prior; wrapped voxels fall from 0.4% to 0.11%.

These are simulated-data results on one anatomy; they validate the operator and the forward physics, not yet the reconstruction on in-vivo data.

1.8 Assumptions and limits

- The concomitant model is first order in G/B_0 (Equation (3)) and uses the *nominal* (or GIRF-filtered) waveforms; eddy-current and heating residuals break the exact A_s-linear structure of spoke-wise coefficients and have not been measured.
- The per-sample correction is a Taylor series in the single phase $\boldsymbol{r}^\top \Delta C \boldsymbol{r}$ about the FOV centre: it is controlled for $\lesssim 0.2$ rad (first order, error $\sim \phi^2/2$ at the edge), needs second order for 0.4–0.7 rad, and a wider generalised stencil beyond. How large the residual is is a sequence-design property: for rotated planar spirals the anisotropic part is ≈ 0.7 of the isotropic one (0.7 rad at the end of the readout), for a yarnball with polar wobble 0.14–0.25 (0.22 rad at 12.7 ms).
- Segments scale linearly with readout length ($L \approx 2\Delta f T_{\text{ro}} + 4$); long readouts and poor shimming pay directly in FFTs.
- The Weyl expansions are exact for quadratic phases only; B0 and GNL enter through the segment diagonals (exact for any spatial function) and, in the normal-operator expansions, to first order with an $O(\Delta^3 \partial^3 \psi)$ remainder.
- Operator accuracy measured on typical images is *not* the relevant quantity for a quasi-Newton preconditioned iteration (see the lagged-correction experiment).

References

[1] M. A. Bernstein, X. J. Zhou, J. A. Polzin, K. F. King, A. Ganin, N. J. Pelc, G. H. Glover. Concomitant gradient terms in phase contrast MR: analysis and correction. *Magn. Reson. Med.* 39:300–308, 1998.

[2] D. G. Norris, J. M. S. Hutchison. Concomitant magnetic field gradients and their effects on imaging at low magnetic field strengths. *Magn. Reson. Imaging* 8:33–37, 1990.

[3] B. P. Sutton, D. C. Noll, J. A. Fessler. Fast, iterative image reconstruction for MRI in the presence of field inhomogeneities. *IEEE Trans. Med. Imaging* 22(2):178–188, 2003.

[4] J. A. Fessler, S. Lee, V. T. Olafsson, H. R. Shi, D. C. Noll. Toeplitz-based iterative image reconstruction for MRI with correction for magnetic field inhomogeneity. *IEEE Trans. Signal Process.* 53(9):3393–3402, 2005.

[5] J. A. Fessler, B. P. Sutton. Nonuniform fast Fourier transforms using min-max interpolation. *IEEE Trans. Signal Process.* 51(2):560–574, 2003.

[6] A. H. Barnett, J. Magland, L. af Klinteberg. A parallel nonuniform fast Fourier transform library based on an "exponential of semicircle" kernel. *SIAM J. Sci. Comput.* 41(5):C479–C504, 2019.

[7] P. Escande, P. Weiss. Approximation of integral operators using product-convolution expansions. *J. Math. Imaging Vis.* 58:333–348, 2017.

[8] G. B. Folland. *Harmonic Analysis in Phase Space.* Princeton University Press, 1989.

[9] T. Hong, X. Xu, J. Hu, J. A. Fessler. Provable preconditioned plug-and-play approach for compressed sensing MRI reconstruction. *IEEE Trans. Comput. Imaging*, 2024 (arXiv:2405.03854).